Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Lumar Dairy Farms

Physical address of dairy:

Number and Street	City	County	Zip Code
7215 S Prairie Flower RD	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/08/1987

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X057-X013-X015-XXXX

B. OPERATORS

Borba, Luis Jr.

Operator name: Borba, Luis Jr.

Telephone no.: Landline: ; Cellular: (209) 495-8438

Mailing Address Number and Street	City	State	Zip Code
7215 S Prairie Flower RD	Turlock	CA	95380

This operator is responsible for paying permit fees.

C. OWNERS

Borba, Luis Jr.

Legal owner name: Borba, Luis Jr.

Telephone no.: Landline: ; Cellular: (209) 495-8438

Mailing Address Number and Street	City	State	Zip Code
7215 S Prairie Flower RD	Turlock	CA	95380

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	1,750	250	150	120	180	350
Maximum number	1,800	250	150	120	180	350
Average number	1,750	250	150	120	180	350
Avg live weight (lbs)	1,300	1,400	900	600		

Predominant milk cow breed: Holstein

Average milk production: 70 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 52,945.85 *tons per reporting period*

Total nitrogen from manure: 685,246.17 *lbs per reporting period*

After ammonia losses (30% loss applied): 479,672.32 *lbs per reporting period*

Total phosphorus from manure: 111,635.06 *lbs per reporting period*

Total potassium from manure: 335,141.15 *lbs per reporting period*

Total salt from manure: 881,475.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 40,230,000 *gallons*

Total nitrogen generated: 191,423.62 *lbs*

Total phosphorus generated: 17,416.47 *lbs*

Total potassium generated: 224,153.88 *lbs*

Total salt generated: 1,554,498.26 *lbs*

	33,450,000	*gallons applied*
+	6,780,000	*gallons exported*
-	0	*gallons imported*
=	40,230,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Barn Well	Ground water
Costa Well	Ground water
Road Well	Ground water
Slough	Ground water
TID Lateral 5	Ground water

Source Description	Type
TID Lateral 5 1/2	Ground water
TID Lateral 7	Ground water
Windmill Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
01/01/2023	Corral solids *Williams Place*	680.00 *ton*	Dry-weight	36.9	14,700.00	4,890.00	9,620.00		64.70
05/11/2023	Corral solids *Williams Plce*	1,850.00 *ton*	Dry-weight	42.8	14,600.00	2,500.00	6,200.00		55.60

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
04/01/2023	Solid commercial fertilizer *10-10-10*	22.05 *ton*	Dry-weight	0.1	10.000000	4.400000	8.300000	0.000000
04/01/2023	Solid commercial fertilizer *UN 32*	41.37 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000
05/29/2023	Solid commercial fertilizer *5-11-4*	13.20 *ton*	Dry-weight	0.1	5.000000	4.800000	3.300000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	32,174.59	3,204.39	4,526.97	0.00
Dry manure	43,514.39	9,487.40	21,377.18	1,731,947.92
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	75,688.99	12,691.79	25,904.15	1,731,947.92

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
04/03/2023	Process wastewater	300,000.00 *gal*	784.00	482.00	0.00	0.00	76.50	930.00		6,980

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
05/16/2023	Process wastewater	1,680,000.00 *gal*	784.00	482.00	0.00	0.00	76.50	930.00		6,980
07/13/2023	Process wastewater	1,140,000.00 *gal*	398.00	311.00	0.00	0.00	37.90	540.00		3,420
07/21/2023	Process wastewater	1,020,000.00 *gal*	398.00	311.00	0.00	0.00	37.90	540.00		3,420
09/04/2023	Process wastewater	1,380,000.00 *gal*	398.00	311.00	0.00	0.00	37.90	540.00		3,420
10/10/2023	Process wastewater	1,260,000.00 *gal*	689.00	353.00	0.00	0.00	39.60	567.00		4,600

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	31,956.18	2,800.01	37,280.62	264,730.10
Total exports for all materials	31,956.18	2,800.01	37,280.62	264,730.10

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Almonds	52	52	1	none	X044-X016-X001-XXXX
August	37	37	2	process wastewater	X057-X007-X003-XXXX
Costa	36	36	2	process wastewater	X057-X014-X002-XXXX
Courtland 14-19	170	170	1	manure	X146-0480-X001-XXXX
Courtland 21	50	50	1	manure	X146-0480-X001-XXXX
Courtland 9-13	141	141	1	manure	X146-0020-X035-XXXX X146-0020-X037-XXXX
Dutra	73	73	2	process wastewater	X057-X007-X008-XXXX
Hilmar	38	38	1	none	X057-X007-X003-XXXX
Lagoon	24	24	2	process wastewater	X057-X013-X015-XXXX
Peaches	63	63	1	none	X044-X016-X001-XXXX X044-X016-X002-XXXX
Prairie Flower	76	76	2	process wastewater	X057-X014-X001-XXXX
Russell Rd	177	177	2	manure	X146-0480-X001-XXXX
Silva	119	119	2	manure	X045-X050-X003-XXXX
Sousa	75	75	2	process wastewater	X057-X024-X003-XXXX X057-X024-X004-XXXX
Tell	32	32	1	none	X045-X010-X001-XXXX X057-X014-X007-XXXX
Wilson Rd East	116	116	2	manure	X132-0200-X030-XXXX X132-0200-X031-XXXX
Wilson Rd West	83	83	2	manure	X132-0200-X029-XXXX
Totals for areas that were used for application	1,177	1,177	23		
Totals for areas that were not used for application	185	185	4		
Land application area totals	1,362	1,362	27		

B. CROPS AND HARVESTS

Almonds

Field name: Almonds

Almonds

01/01/1993: Almond Field Run

Crop: Almond Field Run Acres planted: 52 Plant date: 01/01/1993

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	217.00 *ton*	As-is		4.2	18,800.00	1,650.00	24,800.00		19.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	186.00	0.00
Total actual harvest content	4.17	156.91	13.77	206.98	1,551.15

August

Field name: August

10/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	700.00 *ton*	As-is		68.0	3,520.00	810.00	6,450.00		8.77

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.92	133.19	30.65	244.05	1,061.88

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	1,313.00 *ton*	As-is		68.8	3,000.00	590.00	4,580.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	35.49	212.92	41.87	325.06	1,040.75

Costa

Field name: Costa

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	532.00 *ton*	As-is		60.4	6,770.00	1,180.00	7,720.00		8.35

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.78	200.09	34.88	228.17	977.28

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	1,151.00 *ton*	As-is		69.2	3,670.00	410.00	4,100.00		3.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	31.97	234.68	26.22	262.17	689.32

Courtland 14-19

Field name: Courtland 14-19

06/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 170 Plant date: 06/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	3,349.00 *ton*	As-is		76.6	2,410.00	250.00	1,240.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	19.70	94.95	9.85	48.86	479.42

Courtland 21

Field name: Courtland 21

11/09/2022: Oats, hay

Crop: Oats, hay Acres planted: 50 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	204.00 *ton*	As-is		6.6	10,700.00	1,120.00	7,660.00		8.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.50	115.50	17.85	87.15	0.00
Total actual harvest content	4.08	87.31	9.14	62.51	651.63

Courtland 9-13

Field name: Courtland 9-13

06/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 141 Plant date: 06/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	2,770.00 *ton*	As-is		74.0	2,290.00	340.00	1,340.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	19.65	89.98	13.36	52.65	582.29

Dutra

Field name: Dutra

Dutra

10/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 73 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	1,167.00 *ton*	As-is		62.7	4,180.00	890.00	5,930.00		8.36

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.99	133.65	28.46	189.60	997.00

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 73 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	2,168.00 *ton*	As-is		69.2	3,730.00	580.00	4,570.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	29.70	221.55	34.45	271.45	987.90

Hilmar

Field name: Hilmar

11/15/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 38 Plant date: 11/15/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/15/2023	298.00 *ton*	As-is		9.3	26,600.00	1,730.00	21,600.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	7.84	417.20	27.13	338.78	1,650.17

Lagoon

Field name: Lagoon

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 24 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	444.00 *ton*	As-is		69.8	4,200.00	820.00	6,020.00		9.49

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.50	155.40	30.34	222.74	1,060.41

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 24 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	756.00 *ton*	As-is		70.0	3,600.00	460.00	4,100.00		3.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	31.50	226.80	28.98	258.30	567.00

Peaches

Field name: Peaches

01/01/2008: Peach

Crop: Peach Acres planted: 63 Plant date: 01/01/2008

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
07/23/2023	823.00 *ton*	As-is		80.0	3,200.00	1,400.00	3,300.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	94.50	40.50	99.00	0.00
Total actual harvest content	13.06	83.61	36.58	86.22	0.00

Prairie Flower

Field name: Prairie Flower

10/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 76 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	1,242.00 *ton*	As-is		72.2	5,120.00	960.00	6,710.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.34	167.34	31.38	219.31	963.14

05/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 76 Plant date: 05/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	2,477.00 *ton*	As-is		71.0	3,630.00	380.00	3,890.00		2.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	32.59	236.62	24.77	253.57	491.49

Russell Rd

Field name: Russell Rd

11/09/2022: Oats, hay

Crop: Oats, hay Acres planted: 177 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	684.00 *ton*	As-is		7.8	9,630.00	1,220.00	7,960.00		6.93

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.50	115.50	17.85	87.15	0.00
Total actual harvest content	3.86	74.43	9.43	61.52	493.83

Russell Rd

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 177 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	3,380.00 *ton*	As-is		76.6	2,410.00	250.00	1,240.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	19.10	92.04	9.55	47.36	464.72

Silva

Field name: Silva

10/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 119 Plant date: 10/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	1,807.00 *ton*	As-is		59.4	3,130.00	840.00	5,800.00		8.36

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.18	95.06	25.51	176.14	1,030.80

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 119 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	3,355.00 *ton*	As-is		69.0	3,380.00	460.00	2,840.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.19	190.59	25.94	160.14	996.35

Sousa

Field name: Sousa

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 75 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	1,064.00 *ton*	As-is		67.5	3,030.00	650.00	4,650.00		7.56

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.19	85.97	18.44	131.94	697.13

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	2,512.00 *ton*	As-is		72.0	3,300.00	450.00	2,980.00		0.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	33.49	221.06	30.14	199.62	18.76

Tell

Field name: Tell

11/18/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 32 Plant date: 11/18/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/15/2023	212.00 *ton*	As-is		9.0	26,000.00	1,740.00	18,400.00		13.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.63	344.50	23.06	243.80	1,639.82

Wilson Rd East

Field name: Wilson Rd East

11/08/2022: Oats, hay

Crop: Oats, hay Acres planted: 116 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	456.00 *ton*	As-is		8.6	9,910.00	1,370.00	7,300.00		9.29

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.50	115.50	17.85	87.15	0.00
Total actual harvest content	3.93	77.91	10.77	57.39	667.57

06/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 116 Plant date: 06/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	2,445.00 *ton*	As-is		77.4	1,990.00	260.00	1,220.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	21.08	83.89	10.96	51.43	485.88

Wilson Rd West

Field name: Wilson Rd West

11/08/2022: Oats, hay

Crop: Oats, hay Acres planted: 83 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	405.00 *ton*	As-is		7.2	10,200.00	1,460.00	9,390.00		9.36

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.50	115.50	17.85	87.15	0.00
Total actual harvest content	4.88	99.54	14.25	91.64	847.68

Wilson Rd West

06/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 83 Plant date: 06/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	1,556.00 *ton*	As-is		74.0	2,290.00	340.00	1,340.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	18.75	85.86	12.75	50.24	555.66

NUTRIENT BUDGET

A. LAND APPLICATIONS

Almonds - 01/01/1993: Almond Field Run

Field name: Almonds

Crop: Almond Field Run

Plant date: 01/01/1993

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	149.04	4,422,284.00 *gal*
Application event totals		0.00	0.00	0.00	149.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	298.07	8,844,569.00 *gal*
Application event totals		0.00	0.00	0.00	298.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	28.80	0.00	0.00	0.00	
Application event totals		28.80	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	298.07	8,844,569.00 *gal*
Application event totals		0.00	0.00	0.00	298.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	28.80	0.00	0.00	0.00	
Application event totals		28.80	0.00	0.00	0.00	

Almonds - 01/01/1993: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	298.07	8,844,569.00 *gal*
Application event totals		0.00	0.00	0.00	298.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	28.80	0.00	0.00	0.00	
Application event totals		28.80	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	298.07	8,844,569.00 *gal*
Application event totals		0.00	0.00	0.00	298.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	298.07	8,844,569.00 *gal*
Application event totals		0.00	0.00	0.00	298.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	149.04	4,422,569.00 *gal*
Application event totals		0.00	0.00	0.00	149.04	

August - 10/25/2022: Oats, silage-soft dough

Field name: August

Crop: Oats, silage-soft dough

Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

August - 10/25/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	353.50	7,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	353.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.33	9.15	88.91	562.95	600,000.00 *gal*
Application event totals		63.33	9.15	88.91	562.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	116.70	11.39	138.44	1,039.02	660,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	216.03	4,455,000.00 *gal*
Application event totals		116.70	11.39	138.44	1,255.05	

August - 05/31/2023: Corn, silage

Field name: August

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	159.14	15.53	188.78	1,416.85	900,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	392.78	8,100,000.00 *gal*
Application event totals		159.14	15.53	188.78	1,809.62	

August - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-4	Solid commercial fertilizer	3.00	2.88	1.98	0.00	
Application event totals		3.00	2.88	1.98	0.00	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	366.59	7,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	366.59	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.63	6.15	87.69	555.37	720,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	314.22	6,480,000.00 *gal*
Application event totals		64.63	6.15	87.69	869.59	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	301.13	6,210,000.00 *gal*
Application event totals		0.00	0.00	0.00	301.13	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.63	6.15	87.69	555.37	720,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	314.22	6,480,000.00 *gal*
Application event totals		64.63	6.15	87.69	869.59	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	314.22	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	314.22	

August - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	288.04	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	288.04	

Costa - 10/20/2022: Oats, silage-soft dough

Field name: Costa

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	129.49	8.32	90.54	650.91	600,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	269.13	5,400,000.00 *gal*
Application event totals		129.49	8.32	90.54	920.04	

01/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.04	3.76	36.55	231.43	240,000.00 *gal*
Application event totals		26.04	3.76	36.55	231.43	

04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	109.04	10.64	129.35	970.80	600,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	269.13	5,400,000.00 *gal*
Application event totals		109.04	10.64	129.35	1,239.93	

Costa - 05/28/2023: Corn, silage

Costa - 05/28/2023: Corn, silage

Field name: Costa

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	174.47	17.02	206.96	1,553.28	960,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	322.95	6,480,000.00 *gal*
Application event totals		174.47	17.02	206.96	1,876.23	

05/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-4	Solid commercial fertilizer	3.00	2.88	1.98	0.00	
Application event totals		3.00	2.88	1.98	0.00	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	322.95	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	322.95	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.96	6.85	97.64	618.36	780,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	262.40	5,265,000.00 *gal*
Application event totals		71.96	6.85	97.64	880.76	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.50	7.38	105.15	665.93	840,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	282.58	5,670,000.00 *gal*
Application event totals		77.50	7.38	105.15	948.51	

Costa - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	282.58	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	282.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	262.40	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	262.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	262.40	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	262.40	

Courtland 14-19 - 06/28/2023: Corn, silage

Field name: Courtland 14-19

Crop: Corn, silage

Plant date: 06/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	71.74	12.90	2.28	755.31	1,140.00 *ton*
Application event totals		71.74	12.90	2.28	755.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Courtland 21 - 11/09/2022: Oats, hay

Field name: Courtland 21

Crop: Oats, hay

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Settling Basin	Separator solids	102.67	32.22	48.83	1,803.00	370.00 *ton*
Application event totals		102.67	32.22	48.83	1,803.00	

Courtland 9-13 - 06/28/2023: Corn, silage

Field name: Courtland 9-13

Crop: Corn, silage

Plant date: 06/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	72.07	12.97	2.29	758.88	950.00 *ton*
Application event totals		72.07	12.97	2.29	758.88	

07/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Dutra - 10/21/2022: Oats, silage-soft dough

Field name: Dutra

Crop: Oats, silage-soft dough

Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Dutra - 10/21/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	121.33	7.79	84.84	609.89	1,140,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	252.17	10,260,000.00 *gal*
Application event totals		121.33	7.79	84.84	862.06	

04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.53	6.30	76.55	574.50	720,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	159.26	6,480,000.00 *gal*
Application event totals		64.53	6.30	76.55	733.77	

Dutra - 05/30/2023: Corn, silage

Field name: Dutra

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	129.06	12.59	153.09	1,149.00	1,440,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	318.53	12,960,000.00 *gal*
Application event totals		129.06	12.59	153.09	1,467.53	

05/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-4	Solid commercial fertilizer	3.00	2.88	1.98	0.00	
Application event totals		3.00	2.88	1.98	0.00	

Dutra - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	318.53	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	318.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.52	6.24	88.89	562.98	1,440,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	318.53	12,960,000.00 *gal*
Application event totals		65.52	6.24	88.89	881.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.06	5.72	81.48	516.06	1,320,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	291.98	11,880,000.00 *gal*
Application event totals		60.06	5.72	81.48	808.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	318.53	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	318.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	318.53	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	318.53	

Dutra - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.87	4.94	70.37	445.69	1,140,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	252.17	10,260,000.00 *gal*
Application event totals		51.87	4.94	70.37	697.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	199.08	8,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	199.08	

Hilmar - 11/15/2020: Alfalfa, hay

Field name: Hilmar

Crop: Alfalfa, hay

Plant date: 11/15/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	114.73	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	114.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	229.47	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	229.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	229.47	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	229.47	

Hilmar - 11/15/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	286.83	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	286.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	191.22	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	191.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	210.34	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	210.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	210.34	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	210.34	

Lagoon - 10/20/2022: Oats, silage-soft dough

Field name: Lagoon

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	155.38	9.98	108.65	781.09	480,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	242.21	3,240,000.00 *gal*
Application event totals		155.38	9.98	108.65	1,023.31	

Lagoon - 10/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.25	5.59	67.91	509.67	210,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	141.29	1,890,000.00 *gal*
Application event totals		57.25	5.59	67.91	650.96	

Lagoon - 05/28/2023: Corn, silage

Field name: Lagoon

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	147.21	14.36	174.62	1,310.58	540,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	272.49	3,645,000.00 *gal*
Application event totals		147.21	14.36	174.62	1,583.07	

05/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-4	Solid commercial fertilizer	3.00	2.88	1.98	0.00	
Application event totals		3.00	2.88	1.98	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	242.21	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	242.21	

Lagoon - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.73	7.12	101.39	642.15	540,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	272.49	3,645,000.00 *gal*
Application event totals		74.73	7.12	101.39	914.64	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.82	4.74	67.59	428.10	360,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	181.66	2,430,000.00 *gal*
Application event totals		49.82	4.74	67.59	609.76	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	181.66	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	181.66	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	151.38	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	151.38	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.52	3.95	56.33	356.75	300,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	151.38	2,025,000.00 *gal*
Application event totals		41.52	3.95	56.33	508.13	

Peaches - 01/01/2008: Peach

Field name: Peaches

Crop: Peach

Plant date: 01/01/2008

Peaches - 01/01/2008: Peach

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	257.38	9,252,586.00 *gal*
Application event totals		0.00	0.00	0.00	257.38	

09/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	257.38	9,252,586.00 *gal*
Application event totals		0.00	0.00	0.00	257.38	

10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	128.69	4,626,293.00 *gal*
Application event totals		0.00	0.00	0.00	128.69	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	257.38	9,252,586.00 *gal*
Application event totals		0.00	0.00	0.00	257.38	

04/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
10-10-10	Solid commercial fertilizer	35.00	15.40	29.05	0.00	
Application event totals		35.00	15.40	29.05	0.00	

05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	257.38	9,252,586.00 *gal*
Application event totals		0.00	0.00	0.00	257.38	

Peaches - 01/01/2008: Peach

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
10-10-10	Solid commercial fertilizer	35.00	15.40	29.05	0.00	
Application event totals		35.00	15.40	29.05	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	257.38	9,252,586.00 *gal*
Application event totals		0.00	0.00	0.00	257.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	128.69	4,626,293.00 *gal*
Application event totals		0.00	0.00	0.00	128.69	

Prairie Flower - 10/21/2022: Oats, silage-soft dough

Field name: Prairie Flower

Crop: Oats, silage-soft dough

Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	147.21	9.46	102.93	739.98	1,440,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	305.95	12,960,000.00 *gal*
Application event totals		147.21	9.46	102.93	1,045.94	

Prairie Flower - 10/21/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.48	7.56	91.90	689.78	900,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	191.22	8,100,000.00 *gal*
Application event totals		77.48	7.56	91.90	881.00	

Prairie Flower - 05/29/2023: Corn, silage

Field name: Prairie Flower

Crop: Corn, silage

Plant date: 05/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	134.29	13.10	159.30	1,195.62	1,560,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	248.59	10,530,000.00 *gal*
Application event totals		134.29	13.10	159.30	1,444.21	

05/29/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-4	Solid commercial fertilizer	3.00	2.88	1.98	0.00	
Application event totals		3.00	2.88	1.98	0.00	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.93	5.99	85.38	540.76	1,440,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	229.47	9,720,000.00 *gal*
Application event totals		62.93	5.99	85.38	770.22	

Prairie Flower - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.93	5.99	85.38	540.76	1,440,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	229.47	9,720,000.00 *gal*
Application event totals		62.93	5.99	85.38	770.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.55	6.24	88.94	563.29	1,500,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	239.03	10,125,000.00 *gal*
Application event totals		65.55	6.24	88.94	802.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	219.90	9,315,000.00 *gal*
Application event totals		0.00	0.00	0.00	219.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	239.03	10,125,000.00 *gal*
Application event totals		0.00	0.00	0.00	239.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	305.95	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	305.95	

Russell Rd - 11/09/2022: Oats, hay

Field name: Russell Rd

Crop: Oats, hay

Plant date: 11/09/2022

Russell Rd - 11/09/2022: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Settling Basin	Separator solids	100.33	31.49	47.72	1,761.97	1,280.00 *ton*
Application event totals		100.33	31.49	47.72	1,761.97	

Russell Rd - 06/29/2023: Corn, silage

Field name: Russell Rd

Crop: Corn, silage

Plant date: 06/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	73.13	13.16	2.32	769.98	1,210.00 *ton*
Application event totals		73.13	13.16	2.32	769.98	

07/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slough	Ground water	0.00	0.00	0.00	989.86	23,328,000.00 *gal*
Application event totals		0.00	0.00	0.00	989.86	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slough	Ground water	0.00	0.00	0.00	934.87	22,032,000.00 *gal*
Application event totals		0.00	0.00	0.00	934.87	

Silva - 10/23/2022: Oats, silage-soft dough

Field name: Silva

Crop: Oats, silage-soft dough

Plant date: 10/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	106.01	35.26	69.37	4,665.79	680.00 *ton*
Application event totals		106.01	35.26	69.37	4,665.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 7	Ground water	0.00	0.00	0.00	287.99	21,060,000.00 *gal*
Application event totals		0.00	0.00	0.00	287.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 7	Ground water	0.00	0.00	0.00	129.22	9,450,000.00 *gal*
Application event totals		0.00	0.00	0.00	129.22	

Silva - 06/05/2023: Corn, silage

Field name: Silva

Crop: Corn, silage

Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	259.66	44.46	110.27	9,888.39	1,850.00 *ton*
Application event totals		259.66	44.46	110.27	9,888.39	

Silva - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 7	Ground water	0.00	0.00	0.00	265.83	19,440,000.00 *gal*
Application event totals		0.00	0.00	0.00	265.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-4	Solid commercial fertilizer	3.00	2.88	1.98	0.00	
Application event totals		3.00	2.88	1.98	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 7	Ground water	0.00	0.00	0.00	203.07	14,850,000.00 *gal*
Application event totals		0.00	0.00	0.00	203.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 7	Ground water	0.00	0.00	0.00	232.60	17,010,000.00 *gal*
Application event totals		0.00	0.00	0.00	232.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 7	Ground water	0.00	0.00	0.00	210.45	15,390,000.00 *gal*
Application event totals		0.00	0.00	0.00	210.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 7	Ground water	0.00	0.00	0.00	210.45	15,390,000.00 *gal*
Application event totals		0.00	0.00	0.00	210.45	

Silva - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 7	Ground water	0.00	0.00	0.00	193.84	14,175,000.00 *gal*
Application event totals		0.00	0.00	0.00	193.84	

Sousa - 10/22/2022: Oats, silage-soft dough

Field name: Sousa

Crop: Oats, silage-soft dough

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	387.54	16,200,000.00 *gal*
Application event totals		0.00	0.00	0.00	387.54	

04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	115.15	11.24	136.59	1,025.17	1,320,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	213.15	8,910,000.00 *gal*
Application event totals		115.15	11.24	136.59	1,238.31	

Sousa - 06/02/2023: Corn, silage

Field name: Sousa

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Sousa - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	290.66	12,150,000.00 *gal*
Application event totals		0.00	0.00	0.00	290.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-4	Solid commercial fertilizer	3.00	2.88	1.98	0.00	
Application event totals		3.00	2.88	1.98	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	329.41	13,770,000.00 *gal*
Application event totals		0.00	0.00	0.00	329.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	119.57	11.39	162.23	1,027.44	2,700,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	435.98	18,225,000.00 *gal*
Application event totals		119.57	11.39	162.23	1,463.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	377.85	15,795,000.00 *gal*
Application event totals		0.00	0.00	0.00	377.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.00	8.86	126.18	799.12	2,100,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	339.10	14,175,000.00 *gal*
Application event totals		93.00	8.86	126.18	1,138.22	

Sousa - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	79.71	7.59	108.15	684.96	1,800,000.00 *gal*
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	290.66	12,150,000.00 *gal*
Application event totals		79.71	7.59	108.15	975.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	251.90	10,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	251.90	

Tell - 11/18/2018: Alfalfa, hay

Field name: Tell

Crop: Alfalfa, hay

Plant date: 11/18/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	120.27	2,145,000.00 *gal*
Application event totals		0.00	0.00	0.00	120.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	222.03	3,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	222.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	222.03	3,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	222.03	

Tell - 11/18/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	240.53	4,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	240.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	185.02	3,300,000.00 *gal*
Application event totals		0.00	0.00	0.00	185.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	203.53	3,630,000.00 *gal*
Application event totals		0.00	0.00	0.00	203.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5 1/2	Ground water	0.00	0.00	0.00	185.02	3,300,000.00 *gal*
Application event totals		0.00	0.00	0.00	185.02	

Wilson Rd East - 11/08/2022: Oats, hay

Field name: Wilson Rd East

Crop: Oats, hay

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Settling Basin	Separator solids	101.67	31.90	48.35	1,785.35	850.00 *ton*
Application event totals		101.67	31.90	48.35	1,785.35	

Wilson Rd East - 06/27/2023: Corn, silage

Field name: Wilson Rd East

Crop: Corn, silage

Plant date: 06/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	72.85	13.11	2.31	767.08	790.00 *ton*
Application event totals		72.85	13.11	2.31	767.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Wilson Rd West - 11/08/2022: Oats, hay

Field name: Wilson Rd West

Crop: Oats, hay

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Settling Basin	Separator solids	101.97	32.00	48.50	1,790.67	610.00 *ton*
Application event totals		101.97	32.00	48.50	1,790.67	

Wilson Rd West - 06/27/2023: Corn, silage

Field name: Wilson Rd West

Crop: Corn, silage

Plant date: 06/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Wilson Rd West - 06/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	77.33	13.91	2.45	814.22	600.00 *ton*
Application event totals		77.33	13.91	2.45	814.22	

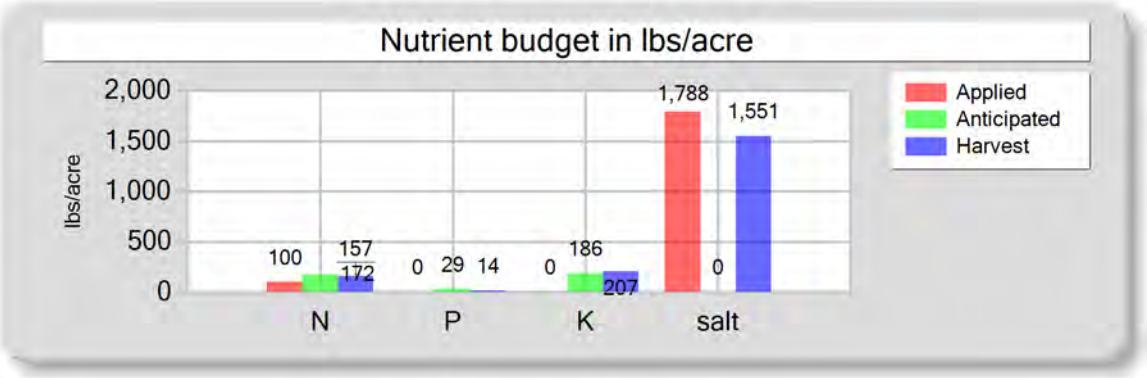
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

B. NUTRIENT BUDGET

Almonds - 01/01/1993: Almond Field Run

Field name: Almonds Crop: Almond Field Run Plant date: 01/01/1993



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	86.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,788.43
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	100.40	0.00	0.00	1,788.43
Anticipated crop nutrient removal	172.00	29.20	186.00	0.00
Actual crop nutrient removal	156.91	13.77	206.98	1,551.15
Nutrient balance	-56.51	-13.77	-206.98	237.28
Applied to removed ratio	0.64	0.00	0.00	1.15

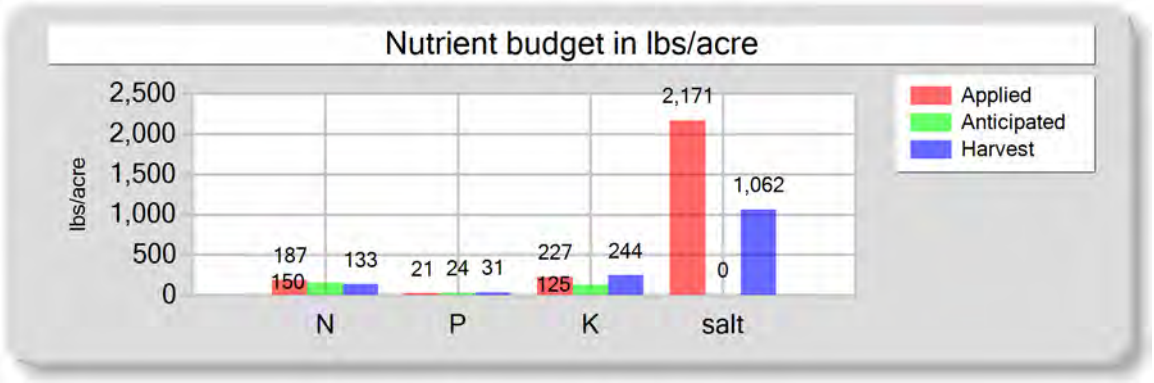
Fresh water applied	
53,067,698.00	*gallons*
1,954.30	*acre-inches*
37.58	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

August - 10/25/2022: Oats, silage-soft dough

Field name: August | Crop: Oats, silage-soft dough | Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	180.04	20.54	227.34	1,601.97
Fresh water	0.00	0.00	0.00	569.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	187.04	20.54	227.34	2,171.50
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	133.19	30.65	244.05	1,061.88
Nutrient balance	53.85	-10.11	-16.71	1,109.62
Applied to removed ratio	1.40	0.67	0.93	2.04

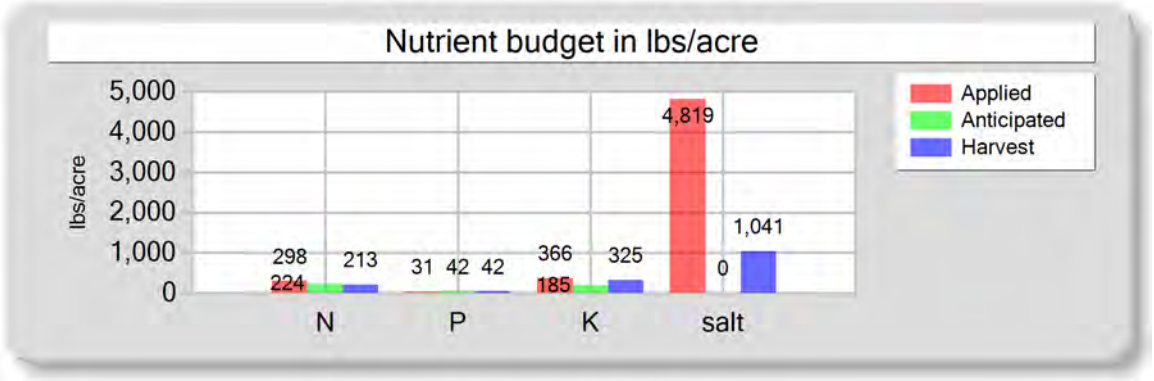
Fresh water applied	
11,745,000.00	*gallons*
432.53	*acre-inches*
11.69	*inches/acre*

Process wastewater applied	
1,260,000.00	*gallons*
46.40	*acre-inches*
1.25	*inches/acre*

Total harvests for the crop	
1	*harvests*

August - 05/31/2023: Corn, silage

Field name: August | Crop: Corn, silage | Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	3.00	2.88	1.98	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	288.40	27.84	364.16	2,527.59
Fresh water	0.00	0.00	0.00	2,291.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.40	30.72	366.14	4,818.80
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	212.92	41.87	325.06	1,040.75
Nutrient balance	85.48	-11.16	41.08	3,778.05
Applied to removed ratio	1.40	0.73	1.13	4.63

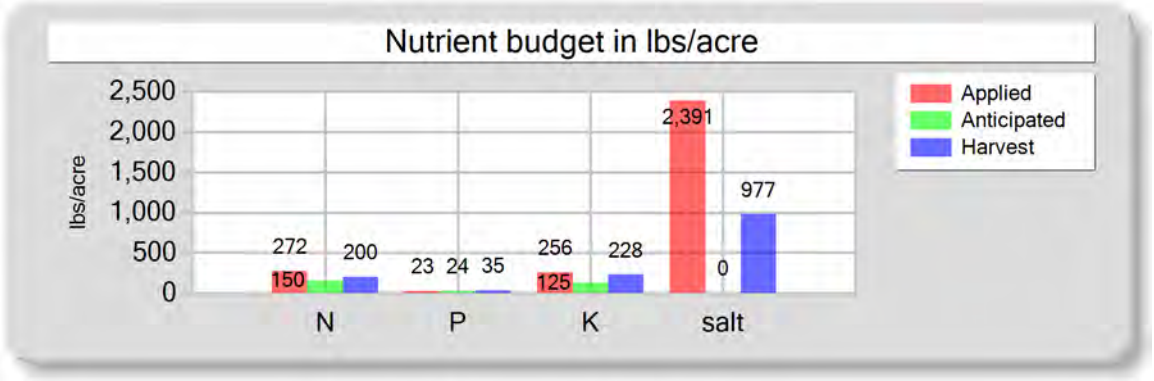
Fresh water applied	
47,250,000.00	*gallons*
1,740.06	*acre-inches*
47.03	*inches/acre*

Process wastewater applied	
2,340,000.00	*gallons*
86.17	*acre-inches*
2.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

Costa - 10/20/2022: Oats, silage-soft dough

Field name: Costa | Crop: Oats, silage-soft dough | Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	264.56	22.72	256.44	1,853.15
Fresh water	0.00	0.00	0.00	538.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	271.56	22.72	256.44	2,391.40
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	200.09	34.88	228.17	977.28
Nutrient balance	71.47	-12.16	28.27	1,414.11
Applied to removed ratio	1.36	0.65	1.12	2.45

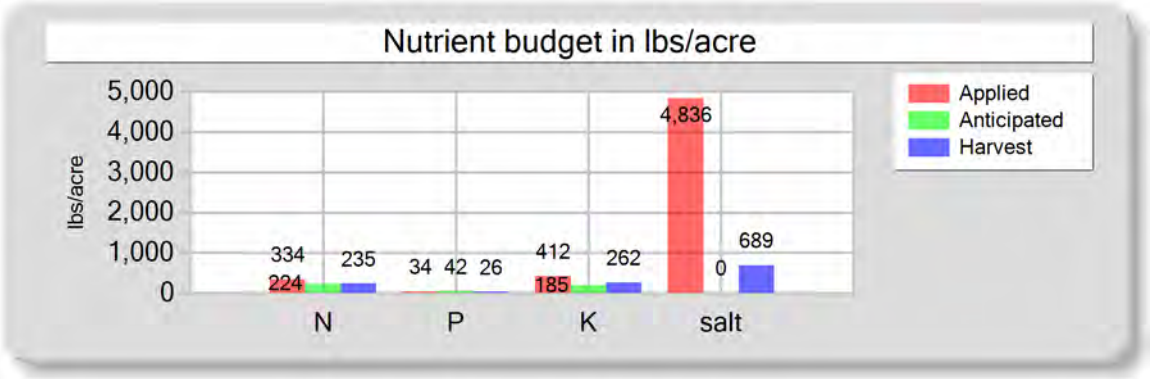
Fresh water applied	
10,800,000.00	*gallons*
397.73	*acre-inches*
11.05	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

Costa - 05/28/2023: Corn, silage

Field name: Costa Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	3.00	2.88	1.98	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	323.93	31.26	409.74	2,837.58
Fresh water	0.00	0.00	0.00	1,998.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	333.93	34.14	411.72	4,835.84
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	234.68	26.22	262.17	689.32
Nutrient balance	99.25	7.92	149.55	4,146.52
Applied to removed ratio	1.42	1.30	1.57	7.02

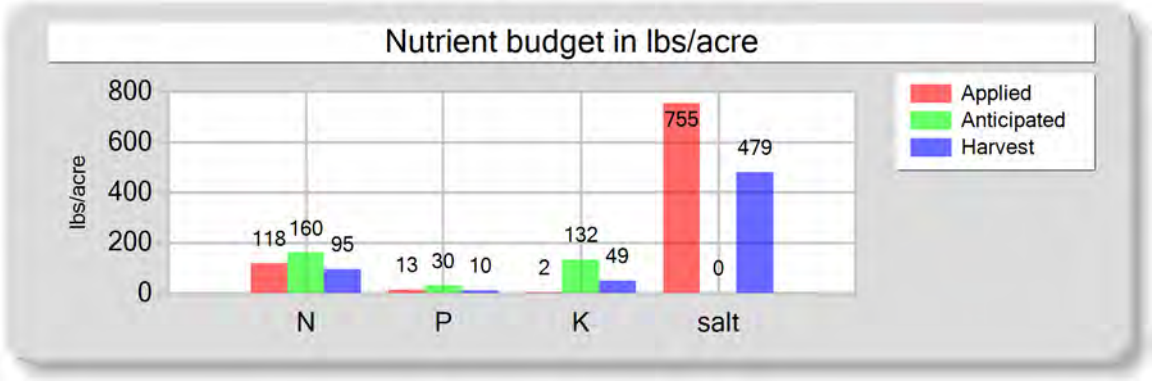
Fresh water applied	
40,095,000.00	*gallons*
1,476.56	*acre-inches*
41.02	*inches/acre*

Process wastewater applied	
2,580,000.00	*gallons*
95.01	*acre-inches*
2.64	*inches/acre*

Total harvests for the crop	
1	*harvests*

Courtland 14-19 - 06/28/2023: Corn, silage

Field name: Courtland 14-19 Crop: Corn, silage Plant date: 06/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	71.74	12.90	2.28	755.31
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	117.74	12.90	2.28	755.31
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	94.95	9.85	48.86	479.42
Nutrient balance	22.78	3.05	-46.58	275.89
Applied to removed ratio	1.24	1.31	0.05	1.58

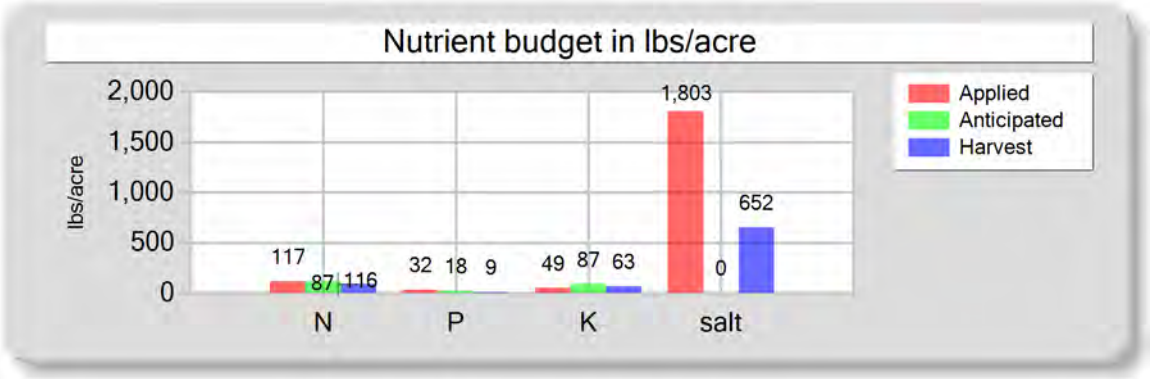
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Courtland 21 - 11/09/2022: Oats, hay

Field name: Courtland 21 Crop: Oats, hay Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	102.67	32.22	48.83	1,803.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	116.67	32.22	48.83	1,803.00
Anticipated crop nutrient removal	115.50	17.85	87.15	0.00
Actual crop nutrient removal	87.31	9.14	62.51	651.63
Nutrient balance	29.36	23.08	-13.67	1,151.36
Applied to removed ratio	1.34	3.53	0.78	2.77

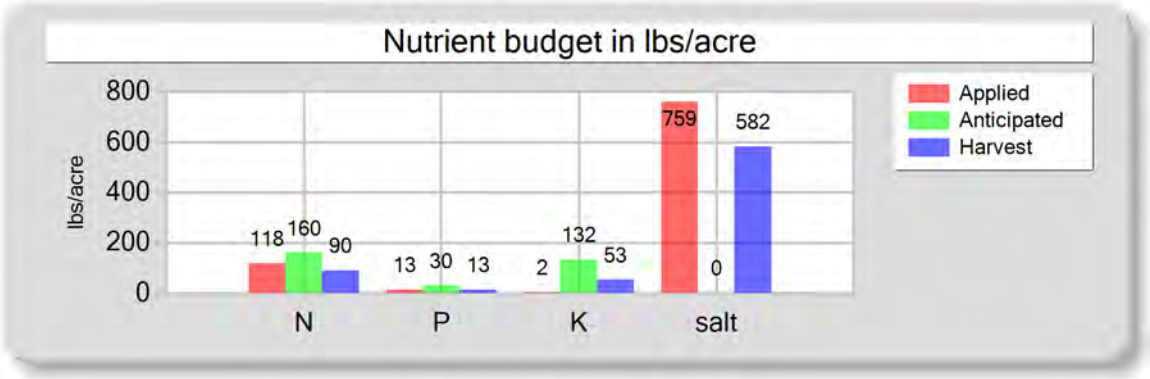
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Courtland 9-13 - 06/28/2023: Corn, silage

Field name: Courtland 9-13 Crop: Corn, silage Plant date: 06/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	72.07	12.97	2.29	758.88
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	118.07	12.97	2.29	758.88
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	89.98	13.36	52.65	582.29
Nutrient balance	28.10	-0.39	-50.36	176.59
Applied to removed ratio	1.31	0.97	0.04	1.30

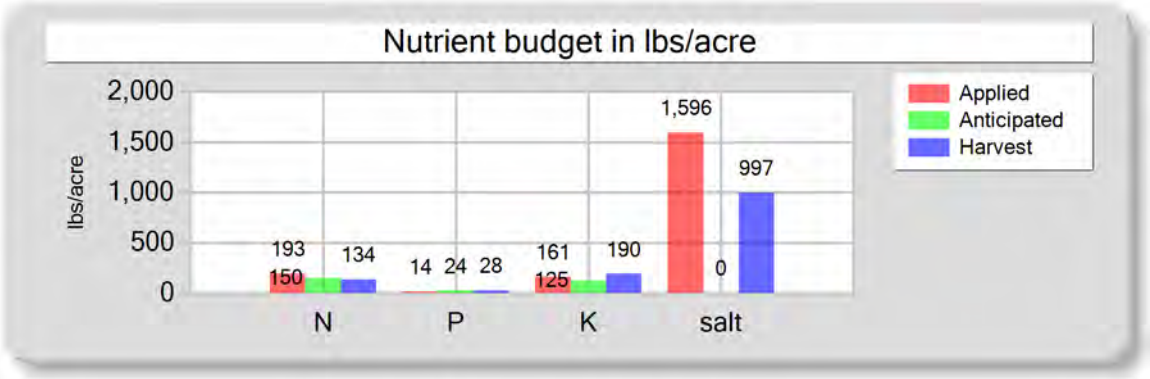
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Dutra - 10/21/2022: Oats, silage-soft dough

Field name: Dutra | Crop: Oats, silage-soft dough | Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	185.86	14.09	161.38	1,184.40
Fresh water	0.00	0.00	0.00	411.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	192.86	14.09	161.38	1,595.83
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	133.65	28.46	189.60	997.00
Nutrient balance	59.21	-14.37	-28.21	598.83
Applied to removed ratio	1.44	0.50	0.85	1.60

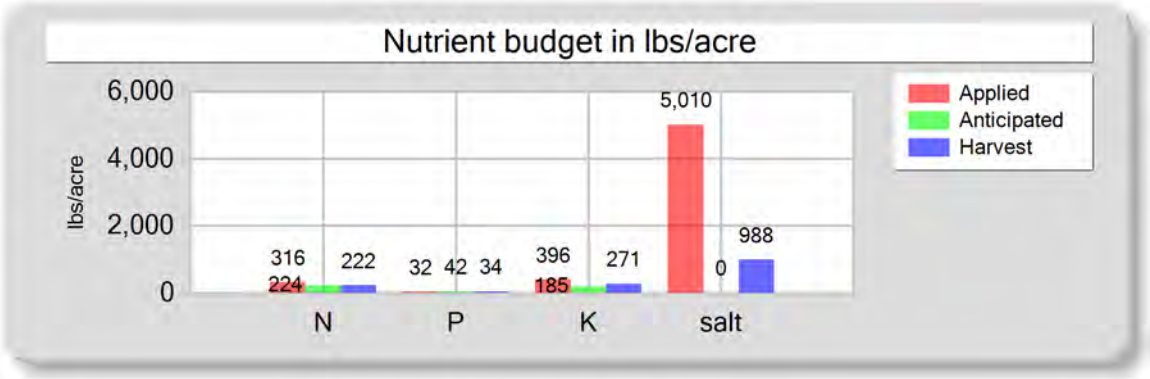
Fresh water applied	
16,740,000.00	*gallons*
616.48	*acre-inches*
8.44	*inches/acre*

Process wastewater applied	
1,860,000.00	*gallons*
68.50	*acre-inches*
0.94	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dutra - 05/30/2023: Corn, silage

Field name: Dutra Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	3.00	2.88	1.98	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	306.50	29.49	393.84	2,673.74
Fresh water	0.00	0.00	0.00	2,335.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	316.50	32.37	395.82	5,009.61
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	221.55	34.45	271.45	987.90
Nutrient balance	94.95	-2.08	124.37	4,021.71
Applied to removed ratio	1.43	0.94	1.46	5.07

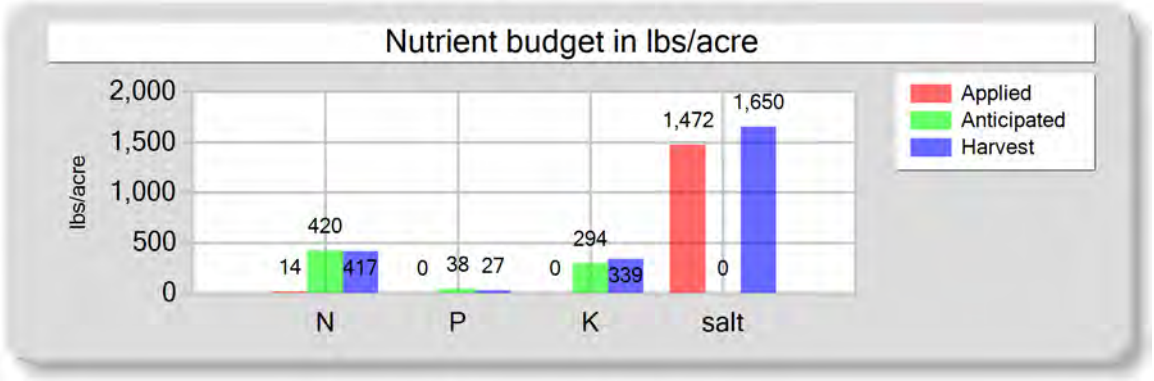
Fresh water applied	
95,040,000.00	*gallons*
3,500.00	*acre-inches*
47.95	*inches/acre*

Process wastewater applied	
5,340,000.00	*gallons*
196.65	*acre-inches*
2.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hilmar - 11/15/2020: Alfalfa, hay

Field name: Hilmar Crop: Alfalfa, hay Plant date: 11/15/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,472.40
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	1,472.40
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	417.20	27.13	338.78	1,650.17
Nutrient balance	-403.20	-27.13	-338.78	-177.76
Applied to removed ratio	0.03	0.00	0.00	0.89

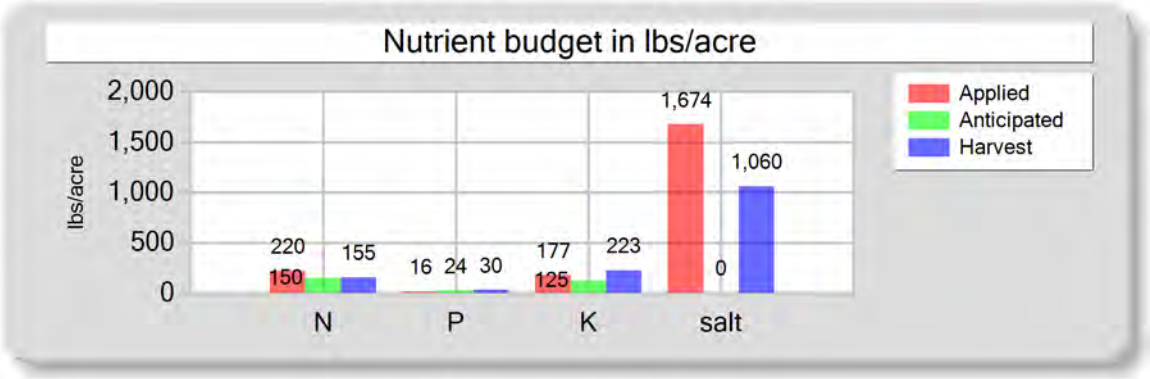
Fresh water applied	
31,185,000.00	*gallons*
1,148.44	*acre-inches*
30.22	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lagoon - 10/20/2022: Oats, silage-soft dough

Field name: Lagoon | Crop: Oats, silage-soft dough | Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.63	15.57	176.56	1,290.76
Fresh water	0.00	0.00	0.00	383.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	219.63	15.57	176.56	1,674.27
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	155.40	30.34	222.74	1,060.41
Nutrient balance	64.23	-14.77	-46.18	613.86
Applied to removed ratio	1.41	0.51	0.79	1.58

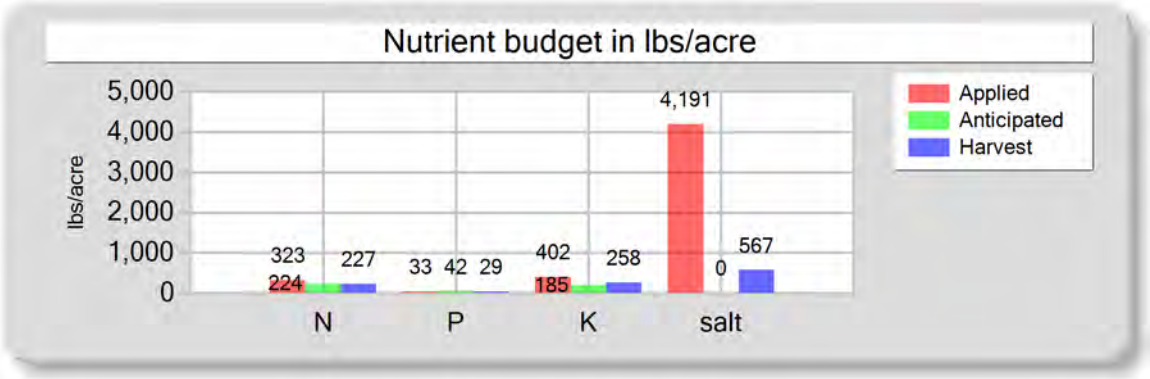
Fresh water applied	
5,130,000.00	*gallons*
188.92	*acre-inches*
7.87	*inches/acre*

Process wastewater applied	
690,000.00	*gallons*
25.41	*acre-inches*
1.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lagoon - 05/28/2023: Corn, silage

Field name: Lagoon Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	3.00	2.88	1.98	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	313.27	30.18	399.93	2,737.58
Fresh water	0.00	0.00	0.00	1,453.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	323.27	33.06	401.91	4,190.86
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	226.80	28.98	258.30	567.00
Nutrient balance	96.47	4.08	143.61	3,623.86
Applied to removed ratio	1.43	1.14	1.56	7.39

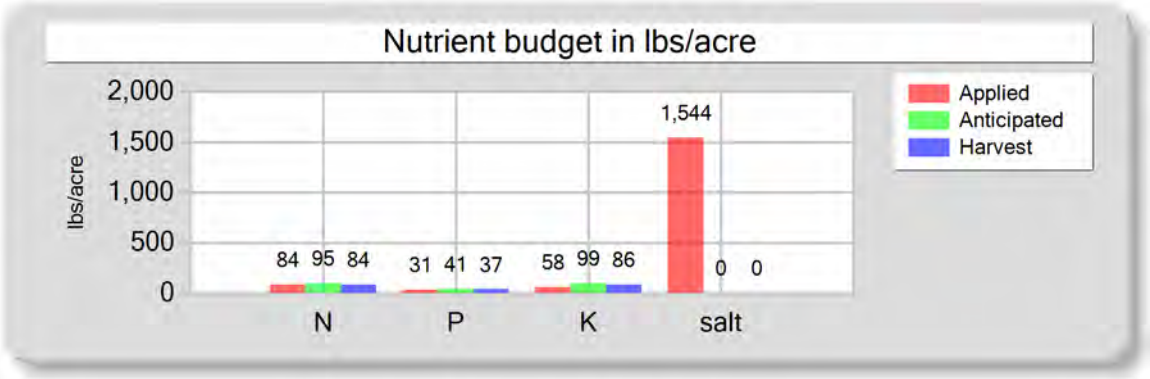
Fresh water applied	
19,440,000.00	*gallons*
715.91	*acre-inches*
29.83	*inches/acre*

Process wastewater applied	
1,740,000.00	*gallons*
64.08	*acre-inches*
2.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

Peaches - 01/01/2008: Peach

Field name: Peaches Crop: Peach Plant date: 01/01/2008



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	70.00	30.80	58.10	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,544.26
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	84.00	30.80	58.10	1,544.26
Anticipated crop nutrient removal	94.50	40.50	99.00	0.00
Actual crop nutrient removal	83.61	36.58	86.22	0.00
Nutrient balance	0.39	-5.78	-28.12	1,544.26
Applied to removed ratio	1.00	0.84	0.67	0.00

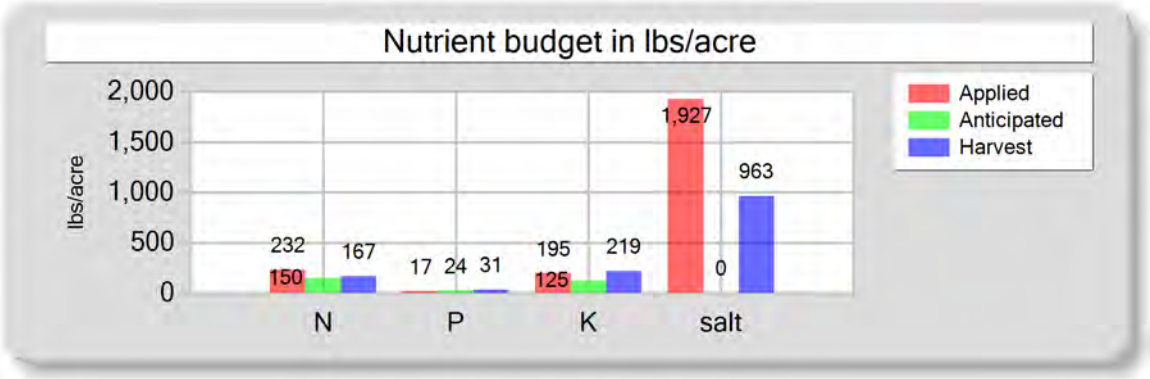
Fresh water applied	
55,515,516.00	*gallons*
2,044.45	*acre-inches*
32.45	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Prairie Flower - 10/21/2022: Oats, silage-soft dough

Field name: Prairie Flower | Crop: Oats, silage-soft dough | Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	224.68	17.02	194.84	1,429.76
Fresh water	0.00	0.00	0.00	497.18
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	231.68	17.02	194.84	1,926.94
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	167.34	31.38	219.31	963.14
Nutrient balance	64.34	-14.36	-24.47	963.80
Applied to removed ratio	1.38	0.54	0.89	2.00

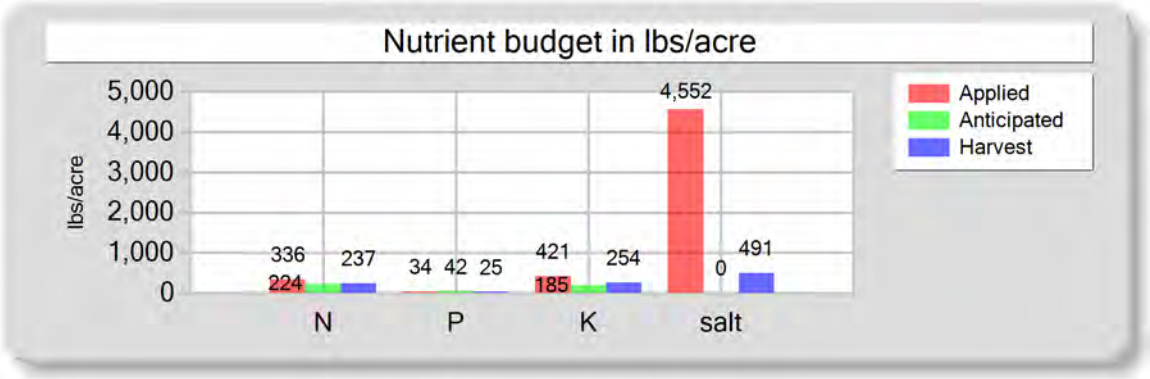
Fresh water applied	
21,060,000.00	*gallons*
775.57	*acre-inches*
10.20	*inches/acre*

Process wastewater applied	
2,340,000.00	*gallons*
86.17	*acre-inches*
1.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

Prairie Flower - 05/29/2023: Corn, silage

Field name: Prairie Flower Crop: Corn, silage Plant date: 05/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	3.00	2.88	1.98	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	325.71	31.33	419.01	2,840.42
Fresh water	0.00	0.00	0.00	1,711.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	335.71	34.21	420.99	4,551.85
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	236.62	24.77	253.57	491.49
Nutrient balance	99.09	9.44	167.42	4,060.36
Applied to removed ratio	1.42	1.38	1.66	9.26

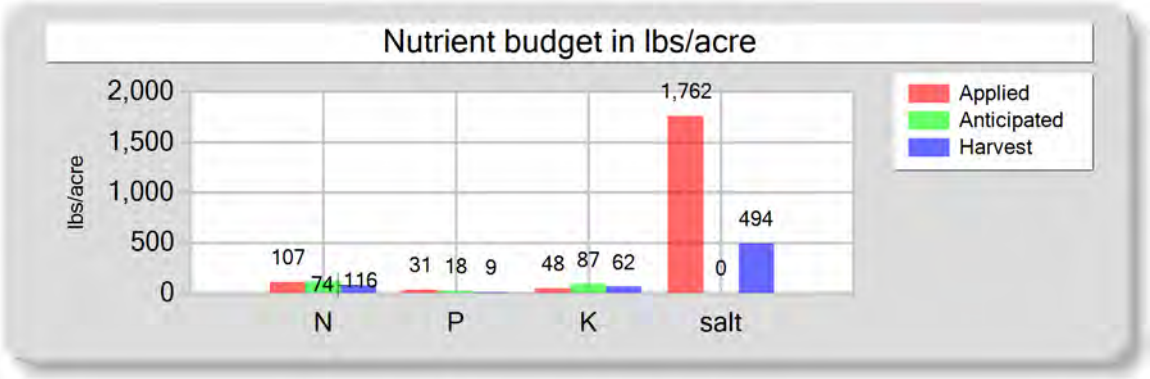
Fresh water applied	
72,495,000.00	*gallons*
2,669.74	*acre-inches*
35.13	*inches/acre*

Process wastewater applied	
5,940,000.00	*gallons*
218.75	*acre-inches*
2.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

Russell Rd - 11/09/2022: Oats, hay

Field name: Russell Rd Crop: Oats, hay Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	100.33	31.49	47.72	1,761.97
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	107.33	31.49	47.72	1,761.97
Anticipated crop nutrient removal	115.50	17.85	87.15	0.00
Actual crop nutrient removal	74.43	9.43	61.52	493.83
Nutrient balance	32.91	22.06	-13.80	1,268.14
Applied to removed ratio	1.44	3.34	0.78	3.57

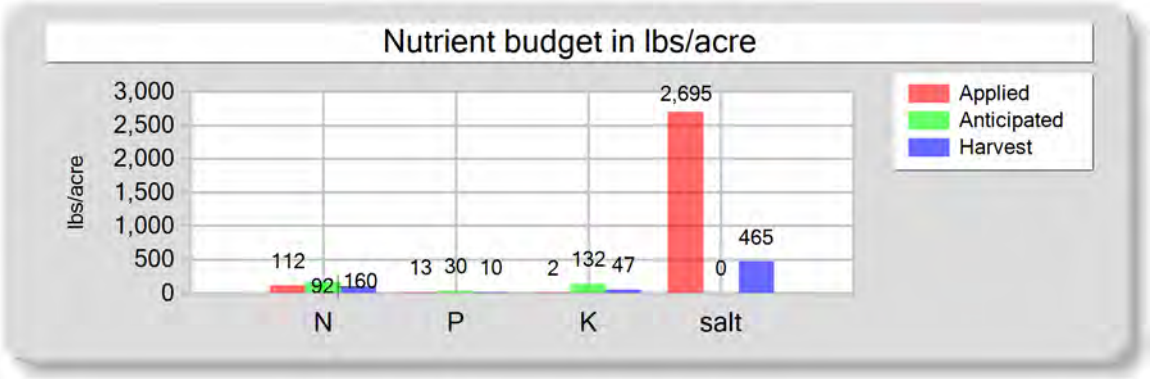
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Russell Rd - 06/29/2023: Corn, silage

Field name: Russell Rd Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	73.13	13.16	2.32	769.98
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,924.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	112.13	13.16	2.32	2,694.71
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	92.04	9.55	47.36	464.72
Nutrient balance	20.09	3.61	-45.04	2,229.99
Applied to removed ratio	1.22	1.38	0.05	5.80

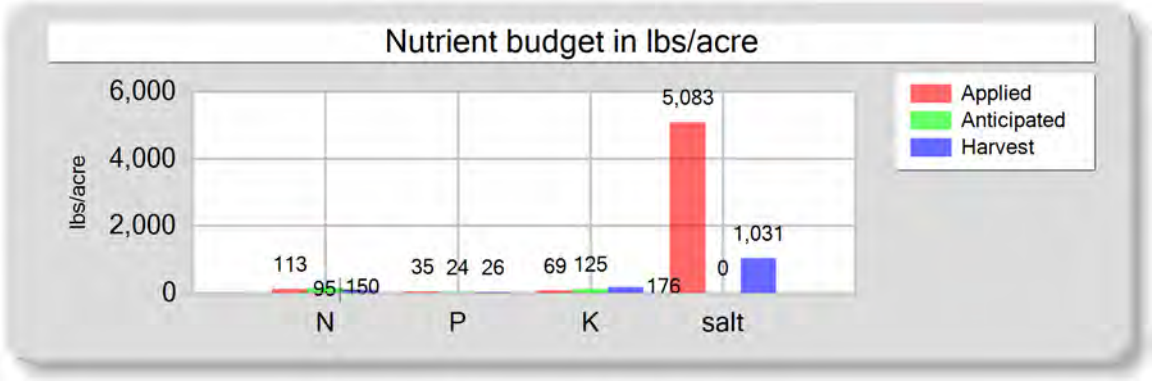
Fresh water applied	
45,360,000.00	*gallons*
1,670.45	*acre-inches*
9.44	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Silva - 10/23/2022: Oats, silage-soft dough

Field name: Silva | Crop: Oats, silage-soft dough | Plant date: 10/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	106.01	35.26	69.37	4,665.79
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	417.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	113.01	35.26	69.37	5,083.01
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	95.06	25.51	176.14	1,030.80
Nutrient balance	17.95	9.75	-106.77	4,052.21
Applied to removed ratio	1.19	1.38	0.39	4.93

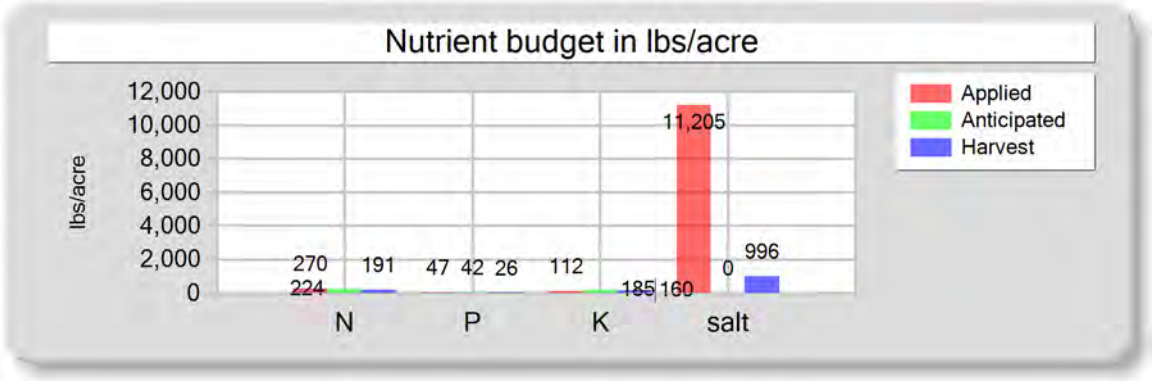
Fresh water applied	
30,510,000.00	*gallons*
1,123.58	*acre-inches*
9.44	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Silva - 06/05/2023: Corn, silage

Field name: Silva | Crop: Corn, silage | Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	3.00	2.88	1.98	0.00
Dry manure	259.66	44.46	110.27	9,888.39
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,316.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	269.66	47.34	112.25	11,204.64
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	190.59	25.94	160.14	996.35
Nutrient balance	79.07	21.40	-47.89	10,208.29
Applied to removed ratio	1.41	1.83	0.70	11.25

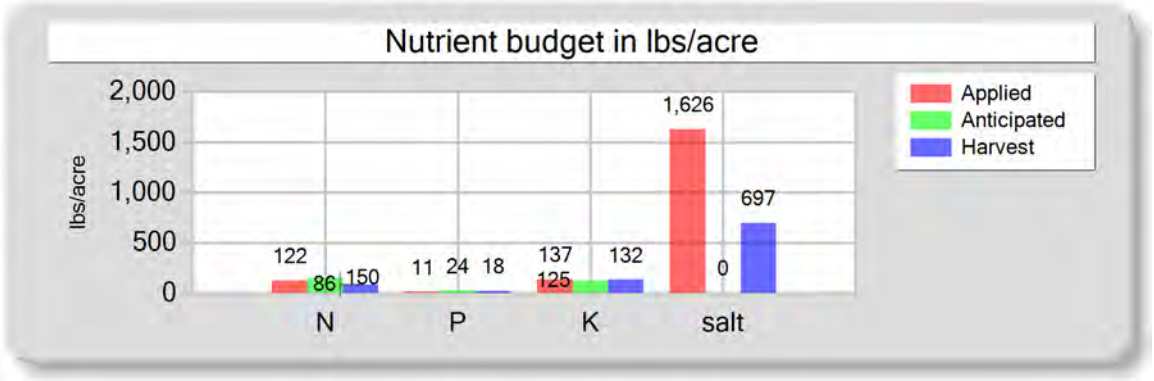
Fresh water applied	
96,255,000.00	*gallons*
3,544.74	*acre-inches*
29.79	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Sousa - 10/22/2022: Oats, silage-soft dough

Field name: Sousa | Crop: Oats, silage-soft dough | Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	115.15	11.24	136.59	1,025.17
Fresh water	0.00	0.00	0.00	600.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	122.15	11.24	136.59	1,625.86
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	85.97	18.44	131.94	697.13
Nutrient balance	36.18	-7.21	4.65	928.72
Applied to removed ratio	1.42	0.61	1.04	2.33

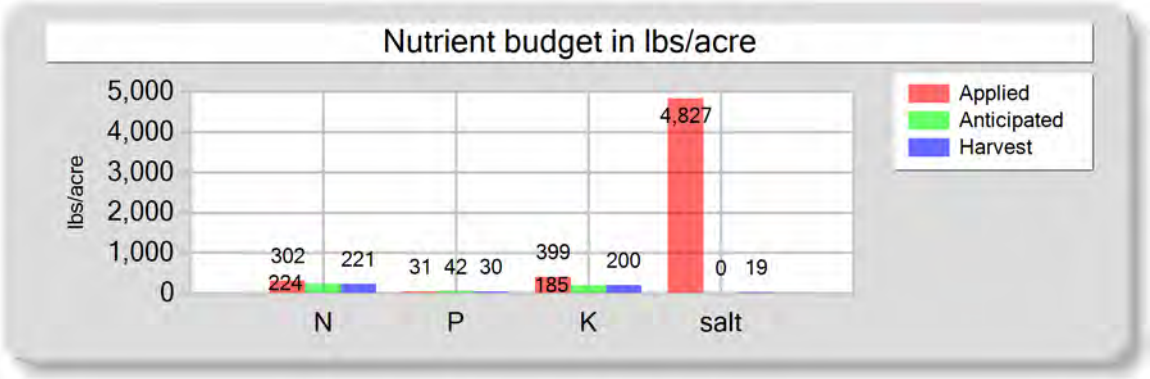
Fresh water applied	
25,110,000.00	*gallons*
924.72	*acre-inches*
12.33	*inches/acre*

Process wastewater applied	
1,320,000.00	*gallons*
48.61	*acre-inches*
0.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

Sousa - 06/02/2023: Corn, silage

Field name: Sousa Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	3.00	2.88	1.98	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	292.28	27.83	396.55	2,511.51
Fresh water	0.00	0.00	0.00	2,315.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	302.28	30.71	398.53	4,827.07
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	221.06	30.14	199.62	18.76
Nutrient balance	81.22	0.57	198.91	4,808.32
Applied to removed ratio	1.37	1.02	2.00	257.36

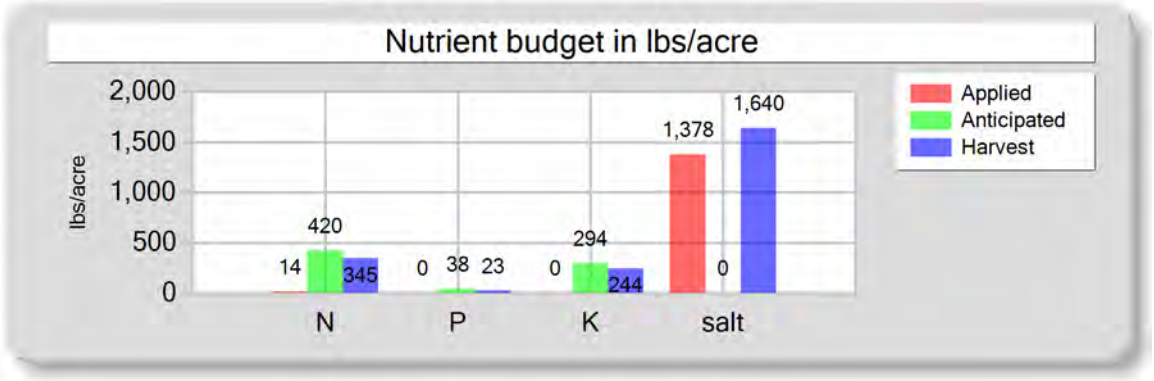
Fresh water applied	
96,795,000.00	*gallons*
3,564.63	*acre-inches*
47.53	*inches/acre*

Process wastewater applied	
6,600,000.00	*gallons*
243.06	*acre-inches*
3.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

Tell - 11/18/2018: Alfalfa, hay

Field name: Tell Crop: Alfalfa, hay Plant date: 11/18/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,378.43
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	1,378.43
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	344.50	23.06	243.80	1,639.82
Nutrient balance	-330.50	-23.06	-243.80	-261.39
Applied to removed ratio	0.04	0.00	0.00	0.84

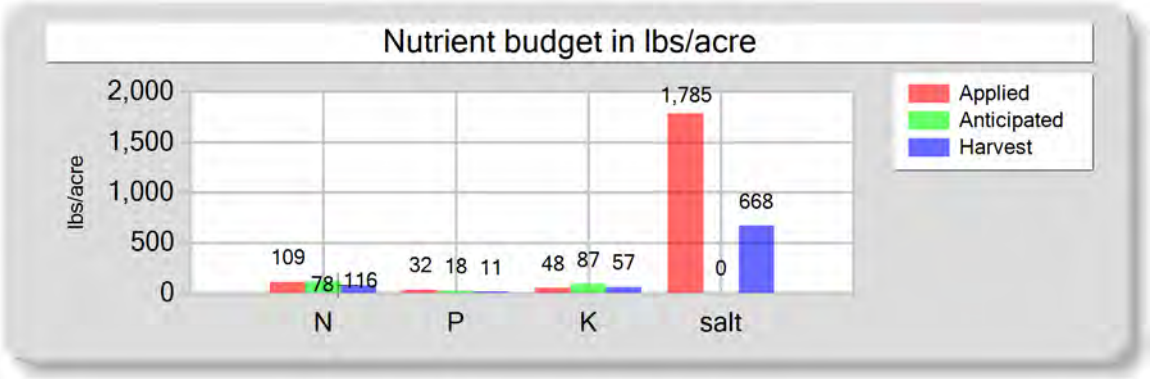
Fresh water applied	
24,585,000.00	*gallons*
905.38	*acre-inches*
28.29	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Wilson Rd East - 11/08/2022: Oats, hay

Field name: Wilson Rd East Crop: Oats, hay Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	101.67	31.90	48.35	1,785.35
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	108.67	31.90	48.35	1,785.35
Anticipated crop nutrient removal	115.50	17.85	87.15	0.00
Actual crop nutrient removal	77.91	10.77	57.39	667.57
Nutrient balance	30.75	21.13	-9.04	1,117.78
Applied to removed ratio	1.39	2.96	0.84	2.67

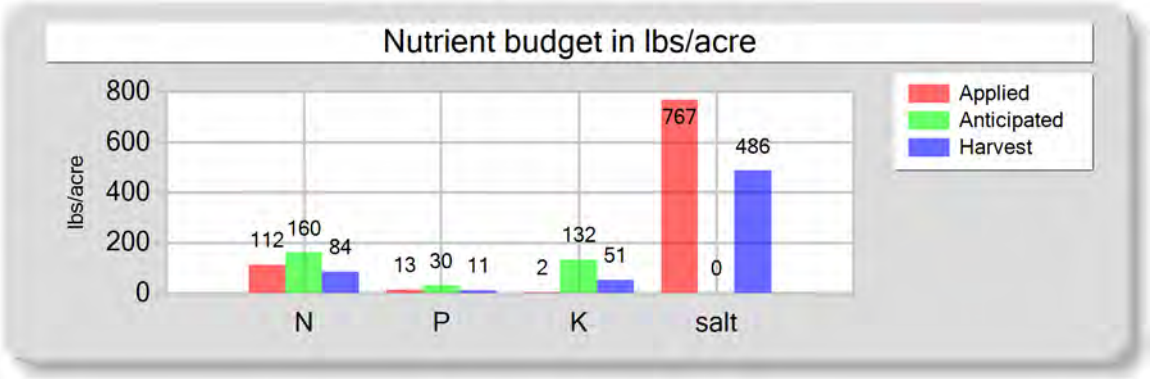
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Wilson Rd East - 06/27/2023: Corn, silage

Field name: Wilson Rd East Crop: Corn, silage Plant date: 06/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	72.85	13.11	2.31	767.08
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	111.85	13.11	2.31	767.08
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	83.89	10.96	51.43	485.88
Nutrient balance	27.96	2.15	-49.12	281.20
Applied to removed ratio	1.33	1.20	0.04	1.58

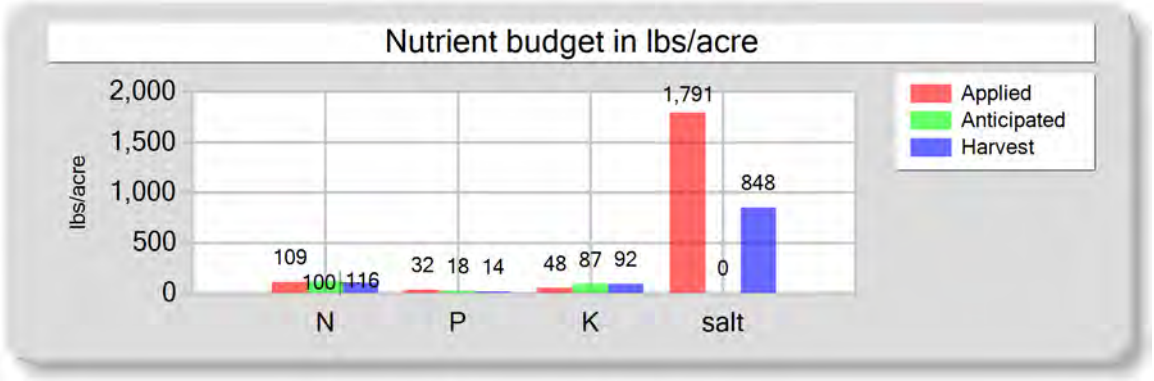
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Wilson Rd West - 11/08/2022: Oats, hay

Field name: Wilson Rd West Crop: Oats, hay Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	101.97	32.00	48.50	1,790.67
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	108.97	32.00	48.50	1,790.67
Anticipated crop nutrient removal	115.50	17.85	87.15	0.00
Actual crop nutrient removal	99.54	14.25	91.64	847.68
Nutrient balance	9.43	17.75	-43.14	942.99
Applied to removed ratio	1.09	2.25	0.53	2.11

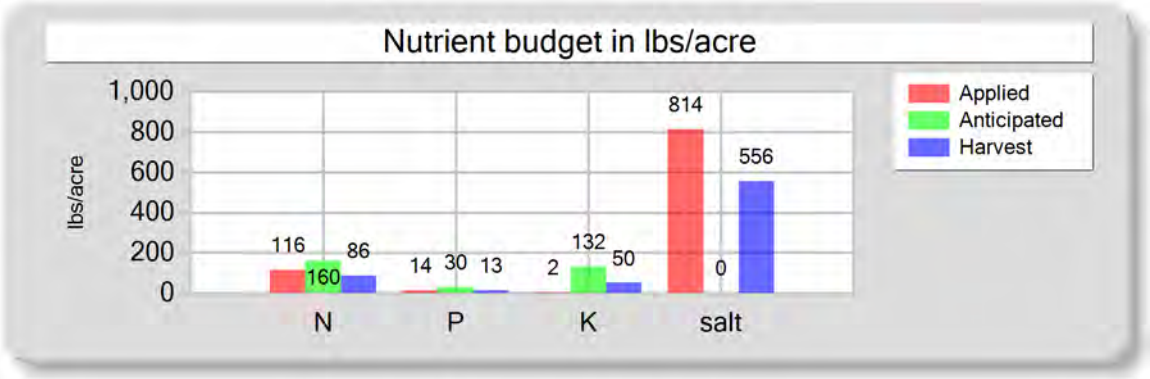
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Wilson Rd West - 06/27/2023: Corn, silage

Field name: Wilson Rd West Crop: Corn, silage Plant date: 06/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	77.33	13.91	2.45	814.22
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	116.33	13.91	2.45	814.22
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	85.86	12.75	50.24	555.66
Nutrient balance	30.47	1.16	-47.79	258.56
Applied to removed ratio	1.35	1.09	0.05	1.47

Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Settling Basin

Sample and source description: Settling Basin

Sample date: 10/13/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,600.00	7,720.00	11,700.00							43.20
DL	100.00	100.00	100.00							0.01

Corral Solids Import

Sample and source description: Corral Solids Import

Sample date: 11/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 36.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	4,890.00	9,620.00	9,490.00	5,520.00	1,390.00	2,500.00	622.00		64.70
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/27/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 4.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	700.00	3,300.00							27.00
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 04/27/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,900.00	3,400.00	600.00							19.90
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 42.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	2,500.00	6,200.00							55.60
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 15.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	3,340.00	10,300.00							46.10
DL	100.00	100.00	100.00							0.01

Settling Basin

Sample and source description: Settling Basin

Sample date: 10/12/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,100.00	5,800.00	8,180.00							42.10
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 10/27/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	931.00	417.00	0.00	0.00	59.80	651.00								8,700.00	4,680
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	468.00	311.00	0.00	0.00	67.60	657.00								6,310.00	4,160
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 06/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	784.00	482.00	0.00	0.00	76.50	930.00								8,800.00	6,980
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	398.00	311.00	0.00	0.00	37.90	540.00								5,900.00	3,420
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	689.00	353.00	0.00	0.00	39.60	567.00								6,120.00	4,600
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Barn Well

Barn Well

Fresh Water

Sample description: Fresh Water

Sample date: 08/24/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	42.10	0.00	42.10								1,730.00	
DL	0.50	0.50	0.50								1.00	

Costa Well

Fresh Water

Sample description: Fresh Water

Sample date: 08/24/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	84.90	0.00	84.90								2,000.00	
DL	0.50	0.50	0.50								1.00	

Road Well

Fresh Water

Sample description: Fresh Water

Sample date: 08/24/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	46.40	0.00	46.40								1,730.00	
DL	0.50	0.50	0.50								1.00	

Slough

Slough

Fresh Water

Sample description: Fresh Water

Sample date: 08/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								1,110.00	900
DL	0.50	0.50	0.50								1.00	10

TID Lateral 5

Fresh Water

Sample description: Fresh Water

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	210
DL	0.50	0.50	0.50								1.00	10

TID Lateral 5 1/2

Fresh Water

Sample description: Fresh Water

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	215
DL	0.50	0.50	0.50								1.00	10

TID Lateral 7

TID Lateral 7

Fresh Water

Sample description: Fresh Water

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	195
DL	0.50	0.50	0.50								1.00	10

Windmill Well

Fresh Water

Sample description: Fresh Water

Sample date: 08/24/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	51.10	0.00	51.10								1,790.00	
DL	0.50	0.50	0.50								1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Almonds - 01/01/1993: Almond Field Run

Almonds - 01/01/1993: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 4.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,800.00	1,650.00	24,800.00		19.40
DL	100.00	100.00	100.00		0.01

Almonds

Sample and source description: Almonds

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 4.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,800.00	1,650.00	24,800.00		19.40
DL	100.00	100.00	100.00		0.01

August - 10/25/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,520.00	810.00	6,450.00		8.77
DL	100.00	100.00	100.00		0.01

August - 05/31/2023: Corn, silage

August - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,000.00	590.00	4,580.00		4.70
DL	100.00	100.00	100.00		0.01

Costa - 10/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,770.00	1,180.00	7,720.00		8.35
DL	100.00	100.00	100.00		0.01

Costa - 05/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,670.00	410.00	4,100.00		3.50
DL	100.00	100.00	100.00		0.01

Courtland 14-19 - 06/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,410.00	250.00	1,240.00		5.20
DL	100.00	100.00	100.00		0.01

Courtland 21 - 11/09/2022: Oats, hay

Forage

Sample and source description: Forage

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 6.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	1,120.00	7,660.00		8.55
DL	100.00	100.00	100.00		0.01

Courtland 9-13 - 06/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,290.00	340.00	1,340.00		5.70
DL	100.00	100.00	100.00		0.01

Dutra - 10/21/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	820.00	5,990.00		8.30
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,180.00	890.00	5,930.00		8.36
DL	100.00	100.00	100.00		0.01

Dutra - 05/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,730.00	580.00	4,570.00		5.40
DL	100.00	100.00	100.00		0.01

Dutra - 05/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,430.00	560.00	4,620.00		5.80
DL	100.00	100.00	100.00		0.01

Hilmar - 11/15/2020: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,600.00	1,730.00	21,600.00		11.60
DL	100.00	100.00	100.00		0.01

Lagoon - 10/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	820.00	6,020.00		9.49
DL	100.00	100.00	100.00		0.01

Lagoon - 05/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	460.00	4,100.00		3.00
DL	100.00	100.00	100.00		0.01

Prairie Flower - 10/21/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,120.00	960.00	6,710.00		10.60
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,150.00	870.00	6,830.00		9.37
DL	100.00	100.00	100.00		0.01

Prairie Flower - 05/29/2023: Corn, silage

Prairie Flower - 05/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,630.00	380.00	3,890.00		2.60
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,460.00	390.00	4,010.00		4.40
DL	100.00	100.00	100.00		0.01

Russell Rd - 11/09/2022: Oats, hay

Forage

Sample and source description: Forage

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 7.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,630.00	1,220.00	7,960.00		6.93
DL	100.00	100.00	100.00		0.01

Russell Rd - 06/29/2023: Corn, silage

Russell Rd - 06/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,410.00	250.00	1,240.00		5.20
DL	100.00	100.00	100.00		0.01

Silva - 10/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,130.00	840.00	5,800.00		8.36
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,010.00	840.00	5,390.00		7.88
DL	100.00	100.00	100.00		0.01

Silva - 06/05/2023: Corn, silage

Silva - 06/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,230.00	510.00	2,990.00		0.10
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,380.00	460.00	2,840.00		5.70
DL	100.00	100.00	100.00		0.01

Sousa - 10/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage
Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 66.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,850.00	680.00	4,910.00		7.18
DL	100.00	100.00	100.00		0.01

Sousa - 10/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,030.00	650.00	4,650.00		7.56
DL	100.00	100.00	100.00		0.01

Sousa - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	333.00	450.00	2,980.00		0.10
DL	100.00	100.00	100.00		0.01

Tell - 11/18/2018: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,000.00	1,740.00	18,400.00		13.60
DL	100.00	100.00	100.00		0.01

Wilson Rd East - 11/08/2022: Oats, hay

Forage

Sample and source description: Forage

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,910.00	1,370.00	7,300.00		9.29
DL	100.00	100.00	100.00		0.01

Wilson Rd East - 06/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	1,990.00	260.00	1,220.00		5.10
DL	100.00	100.00	100.00		0.01

Wilson Rd West - 11/08/2022: Oats, hay

Forage

Sample and source description: Forage

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 7.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	1,460.00	9,390.00		9.36
DL	100.00	100.00	100.00		0.01

Wilson Rd West - 06/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,290.00	340.00	1,340.00		5.70
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

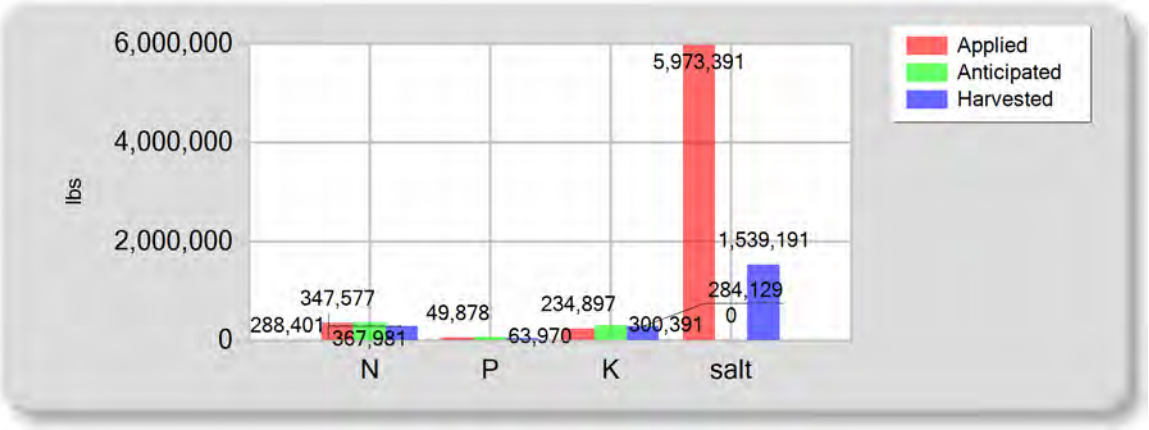
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

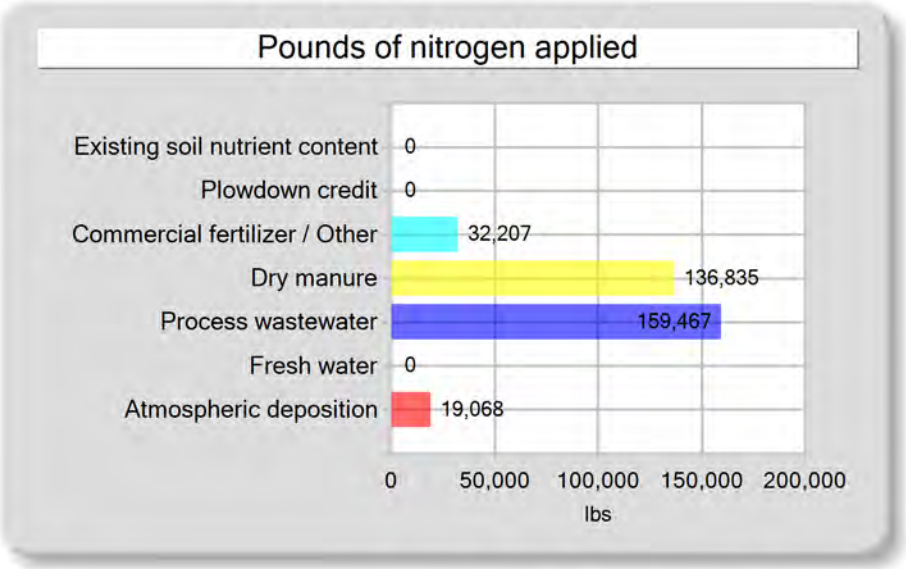
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

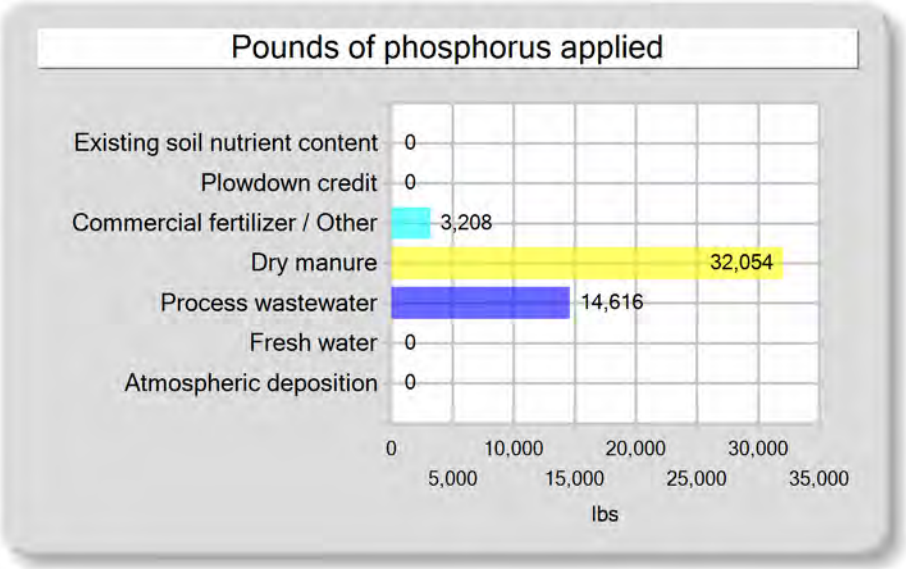
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32,206.80	3,207.60	4,531.50	0.00
Dry manure	136,834.58	32,054.03	43,492.17	3,017,946.66
Process wastewater	159,467.44	14,616.46	186,873.26	1,289,768.15
Fresh water	0.00	0.00	0.00	1,665,676.19
Atmospheric deposition	19,068.00	0.00	0.00	0.00
Total nutrients applied	347,576.82	49,878.09	234,896.93	5,973,391.00
Anticipated crop nutrient removal	367,980.50	63,970.00	300,390.90	0.00
Actual crop nutrient removal	288,400.90	41,856.90	284,129.14	1,539,191.22
Nutrient balance	59,175.92	8,021.19	-49,232.21	4,434,199.78
Applied to removed ratio	1.21	1.19	0.83	3.88

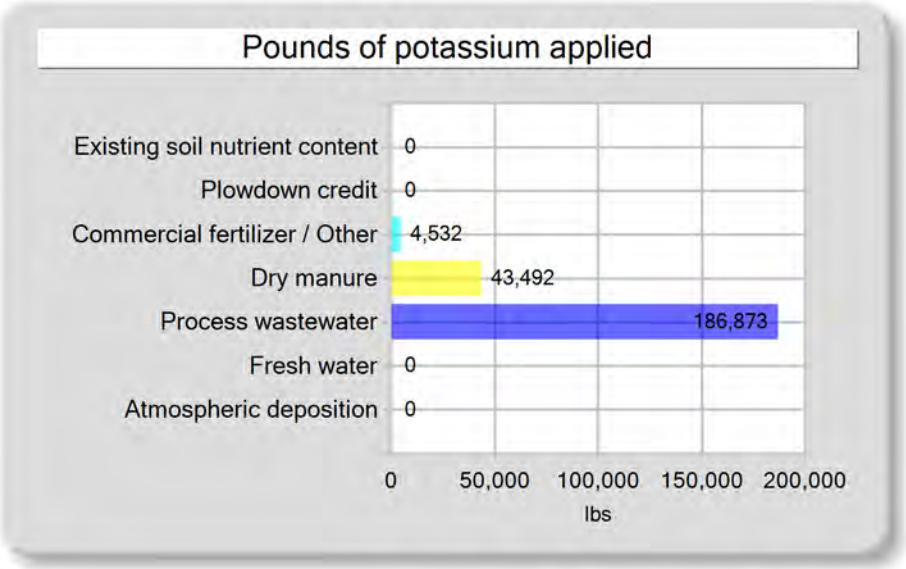
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

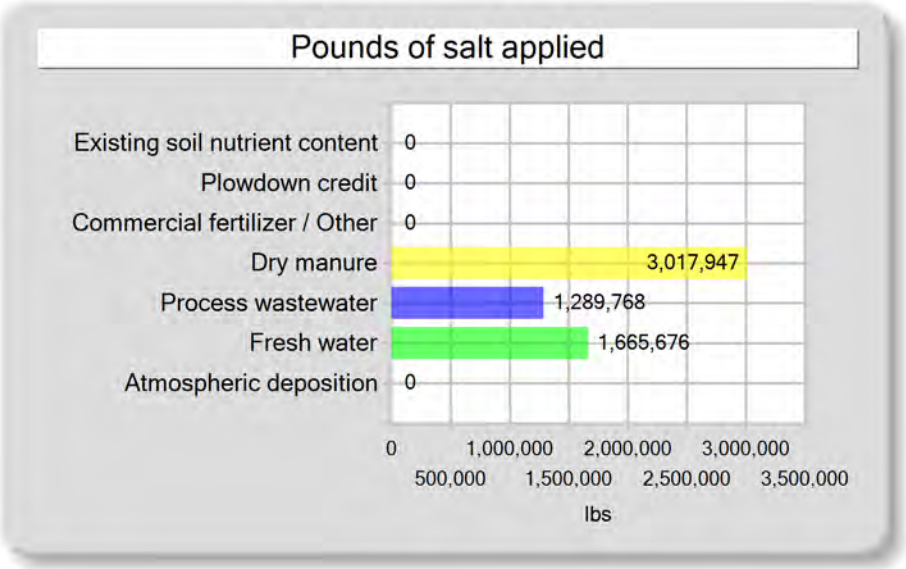


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Luis Borba (Apr 17, 2024 13:51 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Luis Borba, Jr.	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Apr 17, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: LUIS BORBA

Name of Dairy Facility: LUMAIR DAIRY FARMS

Facility Address: 7215 S. PRAIRIE FLOWER RD. TURLOCK 95380
Number and Street City Zip Code

Contact Person Name and Phone Number: LUIS BORBA 209-495-8438
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: BORBA PIPELINE

Address of Hauling Company /Person: _____
Number and Street City Zip Code

Contact Person: _____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

MICKEY BLOUNT 5530 EHRLICH RD. TURLOCK 95380 209-581-2467
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____ 057-013-005
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 4/13/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 300,000 Gallons

Method used to determine volume of process wastewater: _____

5 HOURS @ 1000 GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Luis Borba (Apr 17, 2024 13:51 PDT) Date: Apr 17, 2024

Hauler's Signature: _____ Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: LUIS BORBA

Name of Dairy Facility: LUMAIR DAIRY FARMS

Facility Address: 7215 S. PRAIRIE FLOWER RD. TURLOCK 95380
Number and Street City Zip Code

Contact Person Name and Phone Number: LUIS BORBA 209-495-8438
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: BORBA PIPELINE

Address of Hauling Company /Person: _____
Number and Street City Zip Code

Contact Person: _____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

MICKEY BLOUNT 5530 EHRLICH RD. TURLOCK 95380 209-581-2467
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

057-013-005
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 5/16/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1,680,000 Gallons

Method used to determine volume of process wastewater: _____

28 HOURS @ 1000 GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Luis Borba (Apr 17, 2024 13:51 PDT) Date: Apr 17, 2024

Hauler's Signature: _____ Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: LUIS BORBA

Name of Dairy Facility: LUMAIR DAIRY FARMS

Facility Address: 7215 S. PRAIRIE FLOWER RD. TURLOCK 95380
Number and Street City Zip Code

Contact Person Name and Phone Number: LUIS BORBA 209-495-8438
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: BORBA PIPELINE

Address of Hauling Company /Person: _____
Number and Street City Zip Code

Contact Person: _____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

MICKEY BLOUNT 5530 EHRLICH RD. TURLOCK 95380 209-581-2467
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

057-013-005
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 7/13/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1,140,000 Gallons

Method used to determine volume of process wastewater: _____

19 HOURS @ 1000 GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Luis Borba (Apr 17, 2024 13:51 PDT) Date: Apr 17, 2024

Hauler's Signature: _____ Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: LUIS BORBA

Name of Dairy Facility: LUMAIR DAIRY FARMS

Facility Address: 7215 S. PRAIRIE FLOWER RD. TURLOCK 95380
Number and Street City Zip Code

Contact Person Name and Phone Number: LUIS BORBA 209-495-8438
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: BORBA PIPELINE

Address of Hauling Company /Person: _____
Number and Street City Zip Code

Contact Person: _____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

MICKEY BLOUNT 5530 EHRLICH RD. TURLOCK 95380 209-581-2467
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

057-013-005
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 7/21/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1,020,000 Gallons

Method used to determine volume of process wastewater: _____

17 HOURS @ 1000 GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Luis Borba (Apr 17, 2024 13:51 PDT) Date: Apr 17, 2024

Hauler's Signature: _____ Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: LUIS BORBA

Name of Dairy Facility: LUMAIR DAIRY FARMS

Facility Address: 7215 S. PRAIRIE FLOWER RD. TURLOCK 95380
Number and Street City Zip Code

Contact Person Name and Phone Number: LUIS BORBA 209-495-8438
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: BORBA PIPELINE

Address of Hauling Company /Person: _____
Number and Street City Zip Code

Contact Person: _____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

MICKEY BLOUNT 5530 EHRLICH RD. TURLOCK 95380 209-581-2467
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

057-013-005
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 9/4/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1,380,000 Gallons

Method used to determine volume of process wastewater: _____

23 HOURS @ 1000 GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Luis Borba (Apr 17, 2024 13:51 PDT) Date: Apr 17, 2024

Hauler's Signature: _____ Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: LUIS BORBA

Name of Dairy Facility: LUMAIR DAIRY FARMS

Facility Address: 7215 S. PRAIRIE FLOWER RD. TURLOCK 95380
Number and Street City Zip Code

Contact Person Name and Phone Number: LUIS BORBA 209-495-8438
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: BORBA PIPELINE

Address of Hauling Company /Person: _____
Number and Street City Zip Code

Contact Person: _____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

MICKEY BLOUNT 5530 EHRLICH RD. TURLOCK 95380 209-581-2467
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

057-013-005
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 10/10/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1260,000 Gallons

Method used to determine volume of process wastewater: _____

21 HOURS @ 1000 GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Luis Borba (Apr 17, 2024 13:51 PDT) Date: Apr 17, 2024

Hauler's Signature: _____ Date: _____

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Lumar Dairy Farms
7215 S. Prairie Flower Rd
Turlock, CA 95380

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/24/23 1:12 PM
SUBMITTED BY: Sean
ANALYZED DATE: 8/28/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 23.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Barn Well	T3236042A	Electrical Conductivity (EC)	EPA 120.1 and	1.73	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	42.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1110	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.6	98.2 (T3237014A)	95.8 (T3237014A)	2.47
Electrical Conductivity (EC)	96.3			3.35
Nitrate Nitrogen (NO3-N)	103	97.0 (T3237014A)	97.1 (T3237014A)	0.083

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Lumar Dairy Farms
7215 S. Prairie Flower Rd
Turlock, CA 95380

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/24/23 1:12 PM
SUBMITTED BY: Sean
ANALYZED DATE: 8/28/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 24.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Windmill Well	T3236042B	Electrical Conductivity (EC)	EPA 120.1 and	1.79	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	51.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1150	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.6	98.2 (T3237014A)	95.8 (T3237014A)	2.47
Electrical Conductivity (EC)	96.3			3.35
Nitrate Nitrogen (NO3-N)	103	97.0 (T3237014A)	97.1 (T3237014A)	0.083

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Lumar Dairy Farms
7215 S. Prairie Flower Rd
Turlock, CA 95380

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/24/23 1:12 PM
SUBMITTED BY: Sean
ANALYZED DATE: 8/28/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 25.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Road Well	T3236042C	Electrical Conductivity (EC)	EPA 120.1 and	1.73	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	46.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1110	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.6	98.2 (T3237014A)	95.8 (T3237014A)	2.47
Electrical Conductivity (EC)	96.3			3.35
Nitrate Nitrogen (NO3-N)	103	97.0 (T3237014A)	97.1 (T3237014A)	0.083

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Lumar Dairy Farms
7215 S. Prairie Flower Rd
Turlock, CA 95380

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/24/23 1:12 PM
SUBMITTED BY: Sean
ANALYZED DATE: 8/28/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 23.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Costa Well	T3236042D	Electrical Conductivity (EC)	EPA 120.1 and	2.00	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	84.9	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1280	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.6	98.2 (T3237014A)	95.8 (T3237014A)	2.47
Electrical Conductivity (EC)	96.3			3.35
Nitrate Nitrogen (NO3-N)	103	97.0 (T3237014A)	97.1 (T3237014A)	0.083

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DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: LUMAR DAIRY FARMS
Mailing address:
(not facility address) TURLOCK CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	8/24/23	12:30	BARN WELL	DOMESTIC	23.6°		X				
2		12:35	WINDMILL WELL		24.2°		X				
3		12:40	ROAD WELL		25.0°		X				
4		12:45	COSTA WELL		23.0°		X				
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged In By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	8-24-23 1:12			T3236042

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

Lumar Dairy 2023 Annual Report

Final Audit Report 2024-04-17

Created:	2024-04-17
By:	Sean Locke (slocke@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA9mUulGSZytv03sAqBZMexXpYAXhJsh0h

"Lumar Dairy 2023 Annual Report" History

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